ALCHEMY AOE ALLIANCE

TYPE: METHOD (MTHD)		PROJECT: N/A

MTHD--009	REVISION: OI

Elo Calculation, Final

Table of Contents

1 - Definitions

The following definitions apply:

- g_c – Previously Credited Games: number of games credited to a returning player from the previous instance of a recurring tournament.
- r_A – Previous Actual Win Rate: Number of games a returning player actually won in the previous instance of a recurring tournament, divided by g_c.
- r_T – Previous Theoretical Win Rate: Number of games a returning player should have won in the previous instance of the recurring tournament, divided by g_c. Depends on skill of opponents.
- E_P – Previous Elo: quantified skill for a player from previous participation in a recurring tournament.
- E_T – Tournament Elo: quantified skill calculated from performance within relevant tournaments. Typically this parameter is derived from adjusting an initial seeding elo (from ranked alone) based on wins or losses vs other players.
- E_R – Average Ranked Elo: quantified skill calculated from activity on the ranked ladder, recorded by external APIs. Typically this parameter is weighted for more recent experience, and may be refactored to account for win rate.
- f_T – Tournament Elo Fraction: Value from 0 to 1 weighing the relevance of E_T and E_R to a final skill calculation.
- $\dot{R}$ – Average Ranked Game Rate, in games per time, used to account for the rate at which experience playing ranked games is accumulated.
- $\dot{T}$ – Average Tournament Game Rate, in games per time, used to account for the rate at which experience playing tournament games is accumulated.
- λ – Bias Factor, a dimensionless factor, used to weigh the importance of Ranked Game Rate R over Tournament Game Rate L. Assumed to be equal to 2 unless otherwise specified.

2 - Overview

This document defines a method for adjusting ranked average elo, to factor-in performance from the previous instance of a recurring eSports tournament.

3 - Background

The maturation of Alchemy League demonstrated a need to account for experience gained by participants of a recurring competition. A phenomenon arose, whereby players – gaining valuable experience in Alchemy League – ceased activity on the official ranked ladder, where their newfound skill would manifest. Although the goal of any tournament should be to supplant the ranked ladder as the primary source of entertainment for its players, if the tournament's seeding system depends solely on ranked activity, then such calculations would become increasingly inaccurate. This specification was written to describe how a correction would be performed.

4 - Formulation

4.A - Tournament Elo

Whereas Averaged Ranked Elo E_R reflects player skill under the conditions of the ranked ladder, Tournament Elo E_T is tracked within an isolated pool of elo for a particular tournament. It is a reflection of the types of opponents any one player wins or loses against. An aggregated formula is constructed from terms defined in §1 and presented as Equation 4.A below:

$$E_T = E_p + 32\, g_c (r_A - r_T)$$

Note the resemblance of Equation 4.A to general win/loss elo correction formulae.

4.A.1 - Example

Player1 is seeded with an elo of 1295 in a tournament. Calculate the tournament elo of Player1, if they won 7 of their 8 games, but should have won 4 of 8.

From the problem statement, variables are identified:

Previous Elo (E_P) = 1295

Previously Credited Games (E_R) = 8

Win rates are not directly provided, but can be calculated from the available information:

$$r_A = \frac{7}{8} = 0.875$$

$$r_T = \frac{4}{8} = 0.500$$

Having established the win rates, tournament elo may then be determined from Equation 4.A:

$$E_T = E_p + 32 g_c (r_A - r_T) = 1295 + 32(8)(\frac{7}{8} - \frac{4}{8}) = 1295 + 32(7-4) = 1295 + 96 = 1391$$

Finding the theoretical win rate r_T is non-trivial, and typically if a tournament can claim that a player "should have won X of Y" games, then X was determined from theoretical win rate, and not vice-versa. Such inversion is presented for convenience of this example.

4.B - Final Adjusted Elo

Final Adjusted Elo E_f is a linear combination of E_R and E_T, weighted by f_T to satisfy Equation 4.B below:

$$E_f = f_T E_T + (1 - f_T) E_R$$

4.B.1 - Example

Player2 is a returning league participant with tournament elo of 1800, and average ranked elo of 1600. Calculate the final adjusted elo if tournament elo fraction for Player2 is 0.79.

From the problem statement, variables are identified:

Tournament Elo (E_T) = 1800

Average Ranked Elo (E_R) = 1600

Tournament Elo Fraction (f_T) = 0.79

The remaining step is to calculate Final Adjusted Elo:

$$E_f = f_T E_T + (1 - f_T) E_R = 0.79 * 1800 + (1 - 0.79) * 1600 = 1758$$

4.C - Tournament Elo Fraction

Tournament Elo Fraction f_T is bound to constitutive Equation 4.C below:

$$f_T = \frac{1}{2}[1 - erf(\ln(\frac{\dot{R}}{\lambda \dot{T}}))]$$

Variables on the right hand side of Equation 4.C are defined in §1, “erf” stands for the “error function”, and “ln” is the natural logarithm. The presence of these functions, while complex, assures desirable transition shape between separate elo sources, and Equation 4.C is recommended programming for any elaborate computer system that performs seeding elo adjustments according to this specification. The expression within these transcendental functions may be called the “Rate Quotient” and is denoted by Q, thus $Q = \frac{\dot{R}}{\lambda \dot{T}}$.

For simplicity, Equation 4.C has been plotted in the range of Rate Quotients from 0.1 to 100. The result is presented with semi-logarithmic x-axis as Figure 4.C below:

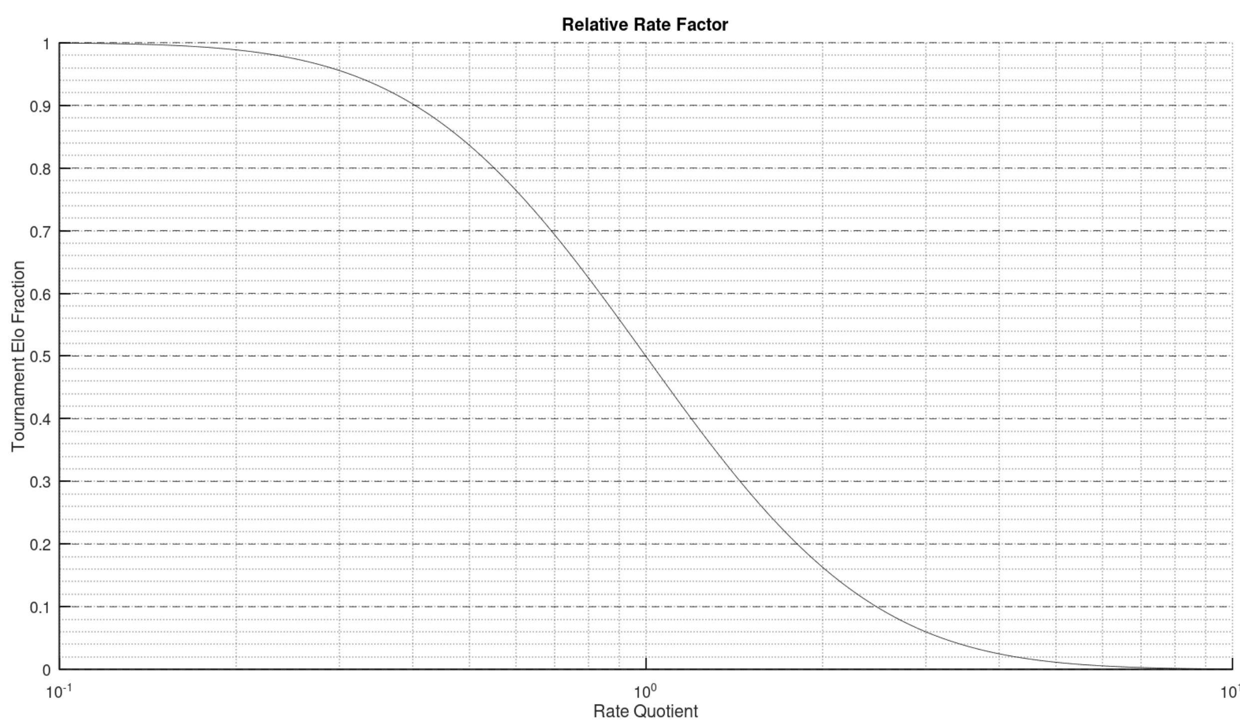

Figure 4.C: Tournament Elo Fraction Curves

Note the symmetry of the result. When $Q \ll 1,\ f_T \to 1$ and when $Q \gg 1,\ f_T \to 0$. Finally, for $Q = 1,\ f_T = \frac{1}{2}$.

4.C.1 - Example

Player3 is a returning league participant with average ranked game rate of 23.4 per season, and 13.9 average league games per season. Calculate Tournament Elo Fraction f_T if the bias was declared as $\lambda = 2.5$.

From the problem statement, variables are identified:

Average Ranked Game Rate ($\dot{R}$) = 23.4

League Game Rate ($\dot{T}$) = 13.9

Bias (λ) = 2.5

The rate quotient may be determined through simple division:

$$Q = \frac{\dot{R}}{\lambda \dot{T}} = \frac{23.4 \frac{games}{season}}{2.5\left(13.9 \frac{games}{season}\right)} = 0.67$$

Figure 4.C.1 is a modification of Figure 4.C. It traces an abscissa at 0.67 Northward from the x-axis until intersecting the curve, and then traces an Ordinate Eastward from that intersection point, until intersecting the y-axis at 0.71.

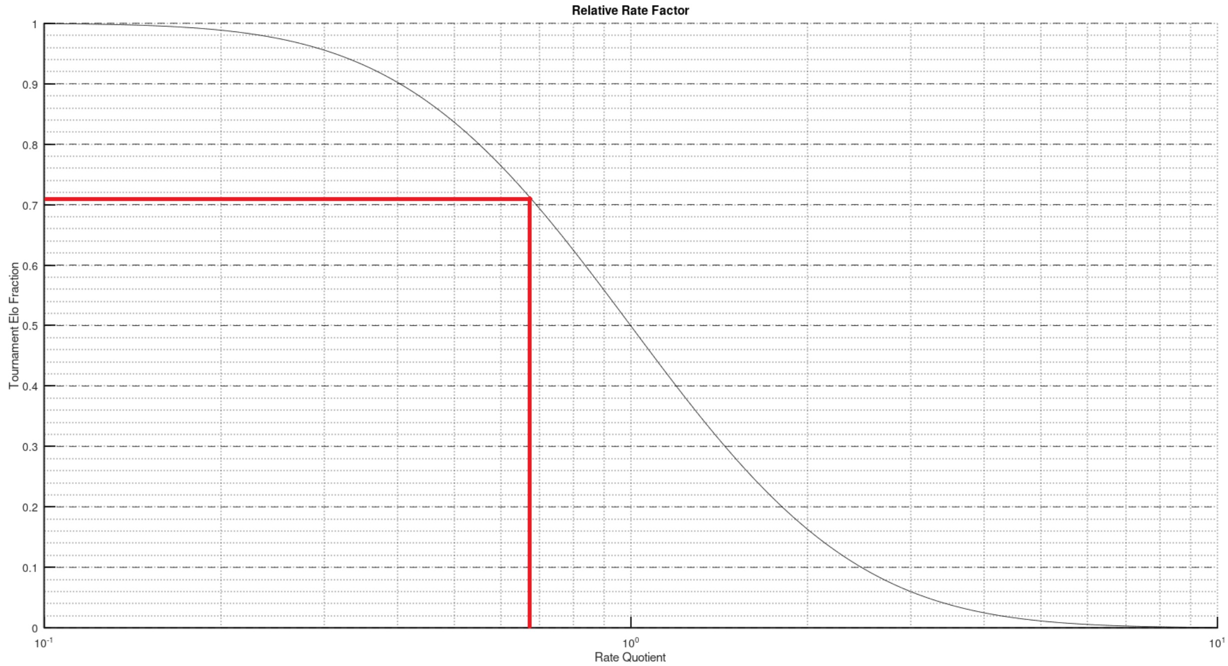

Figure 4.C.1: Tournament Elo Fraction Curve– Simple Case –

Thus, Tournament Elo Fraction $f_T = 0.71$.

4.C.2 - Example

Player4 is a returning participant, seeded at 1588 in the previous Sprint of Alchemy League. Despite being projected to win 42% of all games last Sprint, he won five of his eight games. Player4 also completed 38 League games and 17 Ranked games across 5 Sprints, since first playing in the League. Calculate the new elo of Player4, adjusting for tournament performance, if he is to be currently seeded at 1572, and the bias factor is 1.5.

The problem statement is representative of most tournament scenarios – a final elo is needed for a player who does not play much ranked after settling into their favorite tournament. From the information provided, it may be deduced that this examples combines all those previous in that the following must be determined:

4.C.2.a - Tournament Elo

From the problem statement:

Previous Elo (E_P) = 1588

Previously Credited Games (E_R) = 8

Actual Win Rate (r_A) = 5/8

Theoretical Win Rate (r_A) = 0.42

Invoking Equation 4.A:

$$E_T = E_p + 32 g_c (r_A - r_T) = 1588 + 32(8)\left(\frac{5}{8} - 0.42\right) = 1588 + 32*(5 - 3.36) = 1640.5$$

4.C.2.b - Rate Quotient

From the problem statement:

Average Ranked Game Rate ($\dot{R}$) = 17 games per / 5 Sprints

League Game Rate ($\dot{T}$) = 38 games per / 5 Sprints

Bias $(\lambda) = 1.5$

The rate quotient may be determined through simple division:

$$Q = \frac{\dot{R}}{\lambda \dot{T}} = \frac{\frac{17\, games}{5\, Sprints}}{1.5\left(\frac{38\, games}{5\, Sprints}\right)} = 0.30$$

4.C.2.c - Tournament Elo Fraction

Figure 4.C.2.c is derived from Figure 4.C. A vertical line is placed at Rate Quotient $Q = 0.30$, and then a horizontal line is drawn from where it meets the curve:

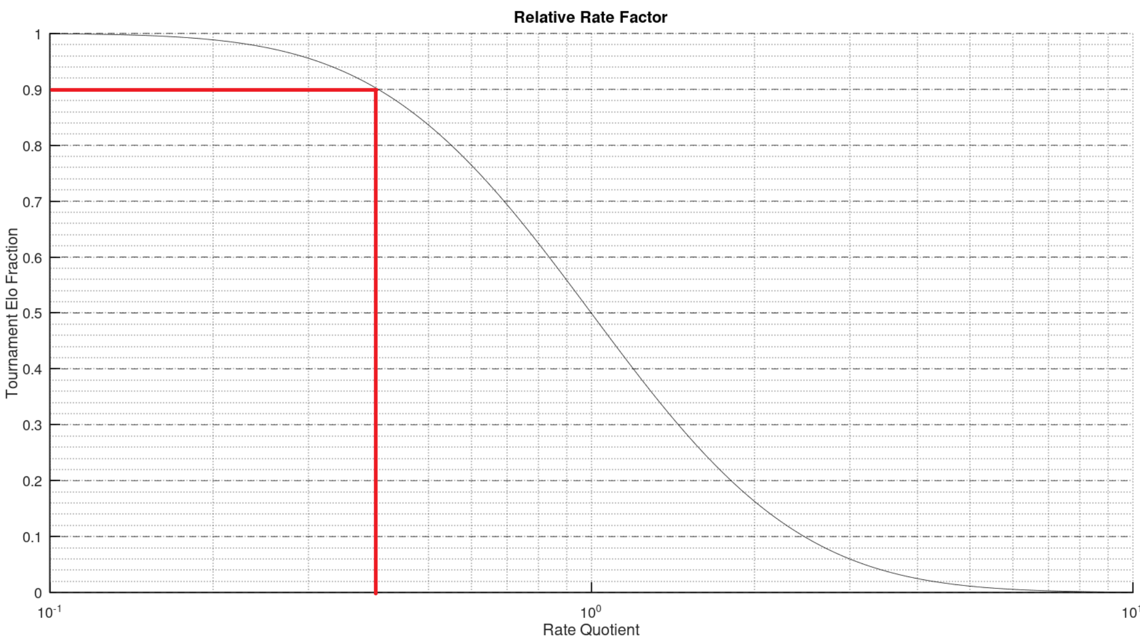

Figure 4.C.2.c: Tournament Elo Fraction Curve – League Case –

Inspection of Figure 4.C.2.c shows that in this situation Player4 has Tournament Elo Fraction $f_T = 0.90$.

4.C.2.d - Final Elo

In §4.C.2.c, it was determined that for the next registration of Player4 in Alchemy League, 90% of his final elo should be sourced from tournament performance, and 10% from the ranked ladder. Thus, variables are:

Tournament Elo $(E_T) =$ 1640.5 (from §4.C.2.a)

Average Ranked Elo (E_R) = 1572 (from §4.C.2)

Tournament Elo Fraction (f_T) = 0.9 (from §4.C.2.c)

Applying Equation 4.B:

$$E_f = f_T E_T + (1-f_T) E_R = 0.9*1640.5+(1-0.9)*1572 = 1633.7$$

Thus, being mostly active in the league, it is logical that Player4's Final Adjusted Elo should be much closer to elo calculated from such activity.

5 - Conclusion

A correction method has been demonstrated, that allows a tournament to account for lack of recent ranked activity among returning players. It incorporates an independently tracked "Tournament Elo" to average against traditional ranked methods, with weighting that depends on Tournament and Ranked game rates. The result is a Final Elo Calculation, encompassing situations where recent tournament experience is regarded as "large" when compared with recent ranked experience.

Signatures:

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